2.3 Group 17

Question Paper

Course	CIE A Level Chemistry
Section	2. Inorganic Chemistry
Topic	2.3 Group 17
Difficulty	Medium

Time allowed: 70

Score: /55

Percentage: /100

Question 1a

A group of students were completing test tube reactions to identify three samples, **X**, **Y** and **Z**, since their labels had fallen off the bottles. All of the samples were solutions of sodium salts.

The positive results of the tests are shown in Table 1.1 below.

Table 1.1

Unknown	Test	Result
X	Added acidified silver nitrate solution Then added concentrated ammonia	Yellow precipitate formed Precipitate remained
Y	Added barium chloride solution	White precipitate formed
Z	Added acidified silver nitrate solution Then added concentrated ammonia	Cream precipitate formed Precipitate dissolved

Using the results from Table 1.1 identify **X**, **Y** and **Z** and write ionic equations, including state symbols, to demonstrate the key reactions taking place in each of the test tube reactions.

- **X**

[2]

- **Y**

[2]

- **Z**

[2]

[6 marks]

Question 1b

Two students are debating the oxidising ability of Group 17.

Student **A** states that the oxidising ability of the halides increases down the group, but Student **B** states that it is the oxidising ability of the halogen molecules that increases down the group.

Is student **A**, student **B** or neither correct? Justify your answer.

[3 marks]

Question 1c

On gently heating, the compound $KClO_3$ reacts as shown in the equation.

$$4KClO_3(s) \rightarrow KCl(s) + 3KClO_4(s)$$

This reaction is an example of disproportionation.

State what is meant by disproportionation and use oxidation numbers to show that disproportionation has taken place.

[3 marks]

Question 1d

Bromine is used to extract iodine from a solution containing iodide ions.

Explain why iodine is a weaker oxidising agent than than bromine.

[3 marks]

Question 2a

The elements of Group 17 of the Periodic Table show trends in their properties.

i)
Complete Table 2.1, stating the colour of each element in its state at room temperature.

[2]

Table 2.1

Halogen	Boiling point / °C	Colour
Chlorine	-36	
Bromine	59	
Iodine	184	

ii)
Briefly explain why the boiling points of the halogens increase from chlorine to iodine.

[2]

[4 marks]

Question 2b

State the full electronic configuration for bromine.

[1 mark]

Question 2c

Explain why bromine is a liquid at room temperature, but fluorine and chlorine are gases.

[3 marks]

Question 2d

The bond enthalpy of the halogen molecules generally decreases down Group 17. Explain why the bond strength of fluorine is lower than that of chlorine.

[3 marks]

Question 3a

Chlorine will react with **cold**, dilute sodium hydroxide.

i)
Write the equation for this reaction. State symbols are not required.

[1]

ii)
State the IUPAC name for the product containing both oxygen and chlorine.

[1]

iii)
Explain why this reaction is an example of a disproportionation reaction.

[2]

[4 marks]

Question 3b

Chlorine is added to drinking water to ensure that it is safe to drink.

i)
State a concern with adding chlorine to drinking water.

[1]

ii)
Write an equation for the reaction of chlorine with cold water.

[1]

[2 marks]

Question 3c

Use an appropriate chemical equation to explain why one of the products formed in the reaction in part **(c)** is classed as a weak acid.

[2 marks]

Question 4a

When solid NaCl reacts with concentrated sulfuric acid the only gas produced is HCl (g) and when solid NaBr reacts with concentrated sulfuric acid the gases produced are brown fumes of Br_2 (g) as well as HBr (g).

Explain the difference in the reactions of concentrated sulfuric acid with NaBr and with NaCl. Your answer should refer to the role of the sulfuric acid in each reaction.

[3 marks]

Question 4b

Solid sodium iodide, NaI (s) reacts with concentrated sulfuric acid to form purple fumes of I_2 (g), a yellow solid and hydrogen sulfide gas. H_2S (g).

i)
Write the half equation for the formation of the yellow solid.

[1]

ii)
Write the half equation for the formation of iodine gas.

[1]

iii)
Hence write the full equation from your answer to parts **(b)(i)** and **(b)(ii)**.

[1]

[3 marks]

Question 4c

Sodium chlorate(V) is a powerful oxidising agent formed by the reaction of chlorine and hot sodium hydroxide is used in the manufacture of matches and as a garden weedkiller.

i)
Write the equation for this reaction. State symbols are not required.

[1]

ii)
Use oxidation numbers to explain why this is a redox reaction.

[1]

iii)
In terms of electrons state the meaning of the term oxidising agent.

[1]

[3 marks]

Question 4d

Chlorine is added to drinking water to ensure that it is safe to drink.

State a concern with adding chlorine to drinking water.

[1 mark]

Question 5a

A group of students are asked to determine the order of reactivity of the halogens and are provided with solutions of potassium chloride, potassium bromide and potassium iodide, as well as aqueous samples of each of the halogens.

i)
Describe how the students would use the reagents named above to determine the trend in reactivity down Group 17 and give expected observations.

ii)
Suggest the trend that the students would find from their results.

iii)
Explain your answer to part **(a)(ii)**, using relevant ionic equations.

[6 marks]

Question 5b

Suggest one extra reagent the students could use to make their observations more clear.

[1 mark]

Question 5c

Halogens react with hydrogen gas to form hydrogen halides. Chlorine reacts with hydrogen when exposed to ultraviolet light to form hydrogen chloride.

i)
Write an equation for the reaction of chlorine and hydrogen.

[1]

ii)
Explain why the thermal stability of the hydrogen halides decreases down the group.

[3]

[4 marks]

Model Answers: Medium

1a

a) Using the results in Table 1.1, the identifies of **X**, **Y** and **Z** are:

- X = sodium iodide / NaI; [1 mark]
- Ag^+ (aq) + I^- (aq) → AgI (s); [1 mark]
- Y = sodium sulfate / Na_2SO_4; [1 mark]
- Ba^{2+} (aq) + SO_4^{2-} (aq) → $BaSO_4$ (s) ; [1 mark]
- Z = sodium bromide / NaBr; [1 mark]
- Ag^+ (aq) + Br^- (aq) → AgBr (s); [1 mark]

[Total: 6 marks]

- If chemical formulae are given rather than names, the formulae must be correct.
- Questions identifying unknown substances are common in the exam
  - You must know all of the tests for ions in detail and be able to provide key equations
- This question tells you that all of the substances contain sodium, but this will not always be the case
  - You need to know the key tests for positive ions as well as for negative ions
- **Always** include state symbols for precipitation reactions, **even if not specified**
  - Sometimes you will not be awarded the mark if you have not included the correct state symbols
  - Sometimes the exam question might not explicitly tell you that you need to include them
  - If you do not need to include them for the mark, they will just be ignored, even if you didn't get them right

1b

b) Is Student **A**, Student **B** or neither student correct?

- Neither student is correct; [1 mark]
- Oxidising ability is the gain of electrons and halides do not gain electrons

OR

Halides lose/donate electrons

OR

Halides reducing ability increases down the group

OR

It is the halogens that gain electrons; [1 mark]

- Oxidising ability of the halogens decreases down the group

  OR

  Oxidising ability of the halogens increases up the group; [1 mark]

[Total: 3 marks]

- You must be clear on the difference between halogens and halides
  - Halogens are the Group 7 elements, which form diatomic molecules (X_2)
  - Halides are ions formed when Group 17 elements gain electrons to become anions (X^-)
- You must also be clear on the difference between oxidising ability and reducing ability
  - Oxidising ability is the ability of a species to oxidise a different species (i.e. how easily it can gain electrons)
  - Reducing ability is the ability of a species to reduce a different species (i.e. how easily it can lose electrons)
- The oxidising ability of the halogens decreases moving down the group as:
  - The atoms get bigger as extra shells of electrons are added, and the shielding increases because of the extra shells
  - The distance between the outer shell and the nucleus increases and the attraction between the nucleus and the outer shell is weaker - it becomes more difficult to gain an electron
- The reducing ability of the halides increases moving down the group
  - This is for the exact same reasons that oxidising ability of halogens decreases

1c

c) Disproportionation is:

- The oxidation and reduction of the same **element**; [1 mark]

Using oxidation numbers we can show disproportionation has taken place as follows:

- Cl is oxidised from +5 (in $KClO_3$) to +7 (in $KClO_4$); [1 mark]
- Cl is reduced from +5 (in $KClO_3$) to -1 (in KCl); [1 mark]

[Total: 3 marks]

- It is important to talk about disproportionation in terms of an element rather than using vague terms such as species
- For example $KClO_3$ is a species but it is only the Cl element in this species that undergoes disproportionation

1d

d) Iodine is a weaker oxidising agent than bromine as:

- Iodine has a larger atomic radius; [1 mark]
- Iodine has greater shielding / more shells; [1 mark]
- Iodine has weaker / less nuclear attraction (on gained electron than bromine); [1 mark]

[Total: 3 marks]

- The reverse argument is allowed for the marks
- Oxidising ability of Group 7 elements decreases as you move down the group for the reasons stated above
- Iodine sits below bromine in the Periodic Table and is therefore able to oxidise other species and itself gain electrons and be reduced

2a

a)

i) Table 2.1 should be completed by:

Halogen	Boiling point / °C	Colour
Chlorine	-36	green / yellow / greenish-yellow
Bromine	59	orange / red / brown
Iodine	184	grey / black

- Chlorine and bromine both correct; [1 mark]
- Iodine correct; [1 mark]

ii) The boiling points of the halogens increase from chlorine to iodine because:

- As you move down the group there are more electrons in the molecule; [1 mark]
- (Hence,) there are stronger instantaneous dipole – induced dipole (id-id) force / London dispersion forces; [1 mark]

[Total: 4 marks]

- You need to know the colours and the trend in the volatility of chlorine, bromine and iodine
  - The colour of the halogens becomes darker down the group
  - They also become less volatile
  - Fluorine is a very pale-coloured gas at room temperature
  - Astatine is a black solid at room temperature

2b

b) The full electronic configuration of bromine is:

- $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^5$
  OR
  $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^5$; **[1 mark]**

[Total: 1 mark]

- You have been asked for the full electronic configuration, therefore you must use $1s^2 2s^2 2p^6 3s^2 3p^6$ and not [Ar] to gain the mark
- Bromine is a period 4 element and is the fifth element in the p block, which means its electronic configuration ends with $4p^5$
  - You then add all the previous electrons to get $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^5$
  - **Remember:** The period 4 d block elements are 3d, not 4d

2c

c) Bromine is a liquid at room temperature but fluorine and chlorine are gases because:

- Bromine has a higher boiling point than fluorine and chlorine; [1 mark]
- Bromine atoms / molecules are bigger
  OR

Bromine atoms have more shells of electrons
OR
Bromine atoms have a bigger atomic radius
OR
There are more electrons in molecules of bromine; [1 mark]

- There are stronger / more intermolecular forces of attraction between bromine molecules / Br_2 (than fluorine / F_2 or chlorine / Cl_2 molecules)
AND
So, more energy is required to pull bromine molecules / Br_2 apart **OR** so, more energy is required to overcome the forces of attraction between bromine molecules / $Br_{2;}$ [1 mark]

[Total: 3 marks]

- The reverse argument for fluorine and chlorine as gases is accepted
- You must be able to state and explain the trend in boiling point down Group 17
- The states change from fluorine and chlorine as gases, to bromine as a liquid and iodine as a solid at room temperature
  - This is evidence that the boiling points increase as you descend the group
- Atoms or molecules which have more electrons, have stronger intermolecular forces of attraction between them
  - Therefore, more heat energy is required to overcome these forces of attraction

2d

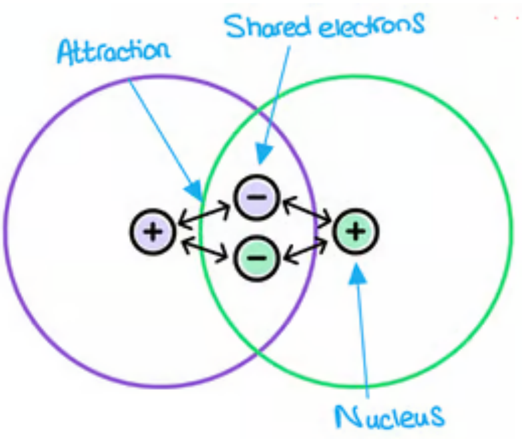
d) Fluorine has a lower bond enthalpy than chlorine because:

- Fluorine is smaller than chlorine so the lone pairs are closer together; [1 mark]
- The lone pairs get so close that they cause repulsion; [1 mark]
- This counteracts the attraction between the bonding pair of electrons and two nuclei; [1 mark]

[Total: 3 marks]

- Bond enthalpy is the heat needed to break one mole of a covalent bond
- The higher the bond enthalpy, the stronger the bond
- The trend down Group 17 is for the bond enthalpy to decrease
  - But, the bond enthalpy for chlorine is greater than that of fluorine due to the

repulsion between the two lone pairs in the molecule
  - In a halogen atom, there are seven electrons in the outermost shell - 3 lone pairs of electrons and one electron that will form the bond
- **Note:** You must say bonding between the nuclei as this is the correct definition of a covalent bond
  - The bonding pair of electrons is attracted to the nuclei on either side and it is this attraction that holds the molecule together



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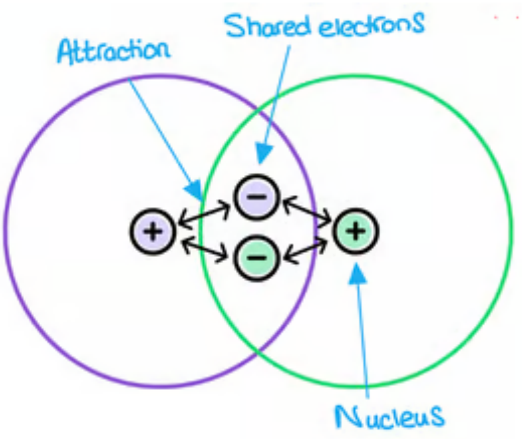

3a

a)

i) The equation for this reaction is:

- $Cl_2 + 2NaOH \rightarrow NaClO + NaCl + H_2O$; [1 mark]

ii) The IUPAC name of the product containing both oxygen and chlorine is:

- Sodium chlorate(I); [1 mark]

iii) It is a disproportionation reaction because:

- The oxidation state of chlorine / Cl changes from 0 as an element / as Cl_2 to +1 in NaClO
  AND
  To -1 in NaCl; [1 mark]
- So, chlorine has been simultaneously oxidised and reduced
  OR
  Chlorine has undergone both oxidation and reduction; [1 mark]

[Total: 4 marks]

- **Careful:** The product in this reaction is sodium chlorate(I) - you need to make sure that you give the correct formula and name (including numeral) in your answer
  - If you do not provide the (I), even if you write sodium chlorate, you will not be awarded the mark
  - Sodium chlorate(V), $NaClO_3$ is a different organic product
- When explaining why this is a disproportionation reaction you must give the starting and finishing oxidation numbers for chlorine
  - Don't just say chlorine is both simultaneously oxidised and reduced
  - You would only gain 1 mark for this answer

3b

b) A concern about adding chlorine to drinking water is:

- Chlorine is potentially toxic if too concentrated
  OR
  Some people may find the taste unpleasant
  OR
  Can react with organic compounds to produce harmful / toxic substances; [1 mark]

The equation for the reaction is:

- $Cl_2 + H_2O \rightarrow HClO + HCl$

OR

$Cl_2 + H_2O \rightarrow 2H^+ + ClO^- + Cl^-$; [1 mark]

[Total: 2 marks]

- You must know all of these reactions of chlorine, as they are asked about frequently in exams
    - Chlorine also reacts with both cold and warm dilute sodium hydroxide to form different products
- You can give a reversible arrow for your equations as well

3c

c) One of the products formed in the reaction in part **(c)** is classed as a weak acid because:

- HClO dissociates / ionises partially; [1 mark]
- $HClO (aq) \rightleftharpoons H^+ (aq) + ClO^- (aq)$; [1 mark]

[Total: 2 marks]

- Both HClO and HCl are formed in the reaction
- HCl is classed as a strong acid as it fully dissociates or ionises
    - In reaction equations, this full dissociation is shown by the use of a standard arrow / $\rightarrow$
    - $HCl \rightarrow H^+ + Cl^-$
- HClO is a weak acid as it dissociates only partially
    - In reaction equations, this partial dissociation is shown by a reversible arrow / $\rightleftharpoons$
    - $HClO (aq) \rightleftharpoons H^+ (aq) + ClO^- (aq)$
- On another note, ClO^- acts as a sterilising agent which cleans the water

4a

a) The difference in the reactions of concentrated sulfuric acid with NaBr and with NaI is:

- Sulfuric acid acts as an acid with NaCl to form HCl fumes

  OR

  $NaCl + H_2SO_4 \rightarrow NaHSO_4 + HCl$

OR
$2NaCl + H_2SO_4 \rightarrow Na_2SO_4 + 2HCl$; [1 mark]

- Sulfuric acid acts as an oxidising agent with NaBr
  OR
  NaBr is a reducing agent; [1 mark]
- Br^- is a more powerful reducing agent than Cl^-
  OR
  Sulfuric acid can oxidise Br^- but not Cl^-
  OR
  Sulfuric acid is a stronger oxidising agent than bromide ions
  OR
  Sulfuric acid is not as strong an oxidising agent as chloride ions; [1 mark]

[Total: 3 marks]

- It gets easier to oxidise the hydrogen halides going down Group 17:
- The halides become stronger reducing agents
- The reaction with NaCl is only an acid-base reaction, not a redox reaction

4b

b)

i) The half equation for the formation of the yellow solid

- $H_2SO_4 + 6H^+ + 6e^- \rightarrow S + 4H_2O$; [1 mark]

ii) The half equation for the formation of iodine gas

- $2I^- \rightarrow I_2 + 2e^-$; [1 mark]

iii) The full equation from your answer to parts **(b)(i)** and **(b)(ii)** is:

- $6I^- + H_2SO_4 + 6H^+ \rightarrow 3I_2 + S + 4H_2O$; [1 mark]

[Total: 3 marks]

- In order to balance $H_2SO_4\ (l) \rightarrow S\ (s)$
- The sulfur atoms are already balanced, so move on to the oxygen atoms by using water
  - $H_2SO_4\ (l) \rightarrow S\ (s) + 4H_2O\ (l)$

- Now balance the hydrogen atoms using hydrogen ions
  - $H_2SO_4 (l) + 6H^+ \rightarrow S (s) + 4H_2O (l)$
- Now balance the charge using electrons
  - $H_2SO_4 (l) + 6H^+ + 6e^- \rightarrow S (s) + 4H_2O (l)$
- As you have been asked to write the full equation, you need to make sure the number of electrons on each half equation is the same, you should multiply the whole equation involving iodine by three
  - $2I^- \rightarrow I_2 + 2e^-$ (x3)
- Having written out both half equations you now need to cancel the electrons
  - $6I^- (g) + H_2SO_4 (l) + 6H^+ + 6e^- \rightarrow 3I_2 (g) + S (s) + 4H_2O (l) + 6e^-$

4c

c)

i) The equation for this reaction:

- $3Cl_2 + 6NaOH \rightarrow 5NaCl + NaClO_3 + 3H_2O$; [1 mark]

ii) To explain why this is a redox reaction using oxidation numbers:

- Chlorine has changed oxidation state from 0 to –1 in NaCl and (+)5 in $NaClO_3$
  AND
  Chlorine has been oxidised and reduced; [1 mark]

iii) The meaning of the term oxidising agent is:

- (A species that) gains electrons
  OR
  An electron donor; [1 mark]

[Total: 3 marks]

- You need to know the equations for the reaction of both hot and cold sodium hydroxide with chlorine
- These reactions form different products
- The oxidation state of chlorine in $NaClO_3$ which is formed by the reaction of hot sodium hydroxide and chlorine has an oxidation number of +5
  - $NaClO_3$
    - Na has an oxidation number of +1

  - Each of the three oxygen atoms has an oxidation number of -2, therefore -6 overall
  - The compound does not have a charge, so the oxidation numbers must all add up to 0
  - $-6 + 1 + x = 0$
  - $x = +5$

4d

d) A concern with adding chlorine to drinking water and an equation for the reaction is:

- Chlorine is potentially toxic if too concentrated
  OR
  Some people may find the taste unpleasant
  OR
  Can react with organic compounds to produce harmful/toxic substances; [1 mark]

4c

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i) The equation for this reaction:

- $3Cl_2 + 6NaOH \rightarrow 5NaCl + NaClO_3 + 3H_2O$; [1 mark]

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- Chlorine is potentially toxic if too concentrated
  OR
  Some people may find the taste unpleasant
  OR
  Can react with organic compounds to produce harmful/toxic substances; [1 mark]

[Total: 1 mark]

5a

a)

i) Students would use displacement reactions to determine the trend in reactivity down Group 7 in the following way:

- React chlorine / bubble chlorine / add chlorine solution into potassium bromide solution
  AND
  The solution would turn an orange / red / brown colour; [1 mark]
- React chlorine with / bubble chlorine / add chlorine solution into potassium iodide solution
  AND
  The solution would turn brown / black colour **OR** a black precipitate / solid is seen; [1 mark]
- React bromine with / add bromine solution into potassium iodide solution
  AND
  The solution would turn brown / black colour **OR** a black precipitate / solid is seen; [1 mark]

ii) The trend that the students would find from their results is:

- Reactivity decreases down Group 17; [1 mark]
- Chlorine is the most reactive / more reactive than bromine and iodine because it displaces bromide and iodide ions
  AND
  Iodide is the least reactive / less reactive than chlorine and bromine because it will not displace chloride or bromide ions; [1 mark]

iii) Using relevant ionic equations:

- $Cl_2 + 2Br^- \rightarrow 2Cl^- + Br_2$
  AND
  $Cl_2 + 2I^- \rightarrow 2Cl^- + I_2$
  AND
  $Br_2 + 2I^- \rightarrow 2Br^- + I_2$; [1 mark]

[Total: 6 marks]

- Halogens will react with metal halides in solution in displacement reactions
  - A more reactive halogen will displace the less reactive halide from its compound
- Questions on this topic are common, especially providing ionic equations
  - Only the halogen and the halide ions are included in the ionic equation
  - The other ions are spectator ions and are not included
- Only ionic equations are allowed, as specified in the question
  - State symbols are not required

5b

b) One extra reagent could be:

- Cyclohexane: [1 mark]

[Total: 1 mark]

- The halogen is more soluble in the organic solvent than the aqueous layer, so the halogen which has been displaced will dissolve in this layer (which would be the top layer)
  - Iodine appears purple
  - Bromine appears orange
  - Chlorine appears pale green

5c

c)

i) The equation for the reaction of chlorine and hydrogen

- $H_2 + Cl_2 \rightarrow 2HCl$; [1 mark]

ii) The thermal stability of the hydrogen halides decreases down the group because:

- The atomic radius of the halogens increases; [1 mark]
- Resulting in a longer bond length; [1 mark]
- The longer the bond, the weaker it is
  AND
  So, less energy is required to break it; [1 mark]

[Total: 4 marks]

- Don't get confused with boiling point when discussing hydrogen halides
  - The boiling points generally decrease down the group
  - **Remember:** This is due to the presence of hydrogen bonding, HF has a higher boiling point than HCl
- This question is asking about bond strength, therefore bond length
  - The longer the bond, the weaker the bond